Week #4: Linear Algebra Foundations

Numerical Methods: Mathematical Foundations

June 22, 2025

Overview

Linear algebra provides the mathematical foundation for a vast array of numerical methods. This week we develop the theoretical framework necessary for understanding numerical linear algebra, focusing on the mathematical structures and properties that determine the behavior of computational algorithms. We explore vector and matrix norms, analyze special classes of matrices, and introduce the singular value decomposition as a unifying tool for understanding matrix properties.

The transition from finite-dimensional linear algebra to numerical computation introduces new considerations. While theoretical linear algebra often assumes exact arithmetic, numerical methods must account for roundoff errors, finite precision, and the computational complexity of operations. This perspective leads us to emphasize concepts such as condition numbers, matrix norms, and stability analysis.

Our approach emphasizes the mathematical theory underlying numerical linear algebra rather than computational procedures. We develop tools for analyzing the sensitivity of linear algebra problems to perturbations, understanding the geometric properties of matrices, and characterizing the fundamental limitations of numerical computation. These theoretical insights provide the foundation for the direct and iterative methods we will study in subsequent weeks.

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1 Vector and Matrix Norms

1.1 Vector Norms

Definition. Vector Norm

A function $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ is a **vector norm** if it satisfies:

1. **Positivity:** $\|x\| \geq 0$ for all $x \in \mathbb{R}^n$, with $\|x\| = 0$ if and only if $x = 0$
2. **Homogeneity:** $\|\alpha x\| = |\alpha|\|x\|$ for all $\alpha \in \mathbb{R}$, $x \in \mathbb{R}^n$
3. **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in \mathbb{R}^n$

Definition. Common Vector Norms

For $x = (x_1, x_2, \ldots, x_n)^T \in \mathbb{R}^n$:

p-norms: $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p\right)^{1/p}$ for $1 \leq p < \infty$

Specific cases:

- $\|x\|_1 = \sum_{i=1}^n |x_i|$ (Manhattan norm)
- $\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$ (Euclidean norm)

- $\|x\|_\infty = \max_{1\le i\le n} |x_i|$ (Maximum norm)

Theorem 1.1 (Norm Equivalence)**.** *All norms on $\mathbb{R}^n$ are equivalent. Specifically, for any two norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ on $\mathbb{R}^n$, there exist positive constants c_1 and c_2 such that:*

$$c_1\|x\|_\alpha \le \|x\|_\beta \le c_2\|x\|_\alpha$$

for all $x \in \mathbb{R}^n$.

Important inequalities:

$$\|x\|_\infty \le \|x\|_2 \le \sqrt{n}\|x\|_\infty \tag{1}$$
$$\|x\|_2 \le \|x\|_1 \le \sqrt{n}\|x\|_2 \tag{2}$$
$$\|x\|_\infty \le \|x\|_1 \le n\|x\|_\infty \tag{3}$$

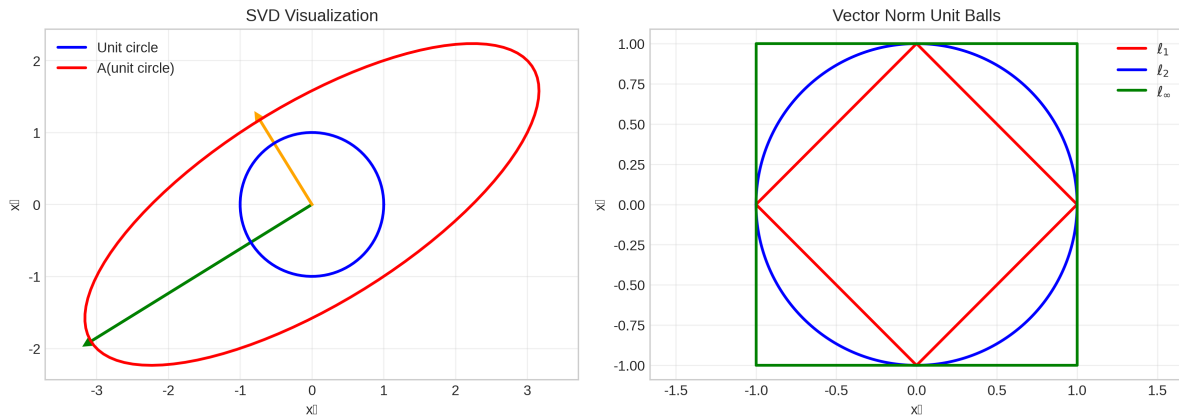

Figure 1: Linear algebra foundations visualization. Top left: Unit balls for different vector norms (ℓ_1, ℓ_2, ℓ_∞) showing their geometric properties. Top right: SVD decomposition components illustrating the geometric interpretation. Bottom left: Matrix condition number effects on linear system sensitivity. Bottom right: Orthogonal matrix properties and their preservation of geometric structure.

1.2 Matrix Norms

Definition. Matrix Norm

A function $\|\cdot\| : \mathbb{R}^{m\times n} \to \mathbb{R}$ is a **matrix norm** if it satisfies the vector norm axioms. Additionally, for square matrices, we often require:

Submultiplicativity: $\|AB\| \le \|A\|\|B\|$ for compatible matrices A and B.

Definition. Induced Matrix Norms

Given a vector norm $\|\cdot\|$ on $\mathbb{R}^n$, the **induced matrix norm** (or operator norm) is:

$$\|A\| = \max_{x\neq 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \|Ax\|$$

Common induced norms:

- $\|A\|_1 = \max_{1\le j\le n} \sum_{i=1}^m |a_{ij}|$ (maximum column sum)
- $\|A\|_\infty = \max_{1\le i\le m} \sum_{j=1}^n |a_{ij}|$ (maximum row sum)

- $\|A\|_2 = \sigma_1(A)$ (largest singular value)

Definition. Frobenius Norm

The **Frobenius norm** is defined as:

$$\|A\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n}|a_{ij}|^2} = \sqrt{\mathrm{tr}(A^TA)} = \sqrt{\sum_{i=1}^{\min(m,n)}\sigma_i^2}$$

where σ_i are the singular values of A.

The Frobenius norm is not an induced norm but satisfies submultiplicativity.

Theorem 1.2 (Properties of Induced Norms)**.** *For an induced matrix norm $\|\cdot\|$:*

1. $\|I\| = 1$ *(where I is the identity matrix)*
2. $\|Ax\| \leq \|A\|\|x\|$ *for all compatible A and x*
3. $\|AB\| \leq \|A\|\|B\|$ *for compatible matrices A and B*
4. *If A is invertible, then* $\|A^{-1}\| \geq 1/\|A\|$

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2 Condition Numbers and Sensitivity Analysis

2.1 Matrix Condition Numbers

Definition. Condition Number

For an invertible matrix $A \in \mathbb{R}^{n\times n}$ and a given matrix norm $\|\cdot\|$, the **condition number** is:

$$\kappa(A) = \|A\|\|A^{-1}\|$$

For the 2-norm: $\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$
where $\sigma_{\max}$ and $\sigma_{\min}$ are the largest and smallest singular values of A.

Theorem 2.1 (Properties of Condition Numbers)**.**

1. $\kappa(A) \geq 1$ *for any invertible matrix A*
2. $\kappa(I) = 1$ *(identity matrix is perfectly conditioned)*
3. $\kappa(\alpha A) = \kappa(A)$ *for any* $\alpha \neq 0$
4. $\kappa(A) = \kappa(A^{-1})$
5. *For orthogonal matrices Q:* $\kappa_2(Q) = 1$
6. $\kappa(AB) \leq \kappa(A)\kappa(B)$

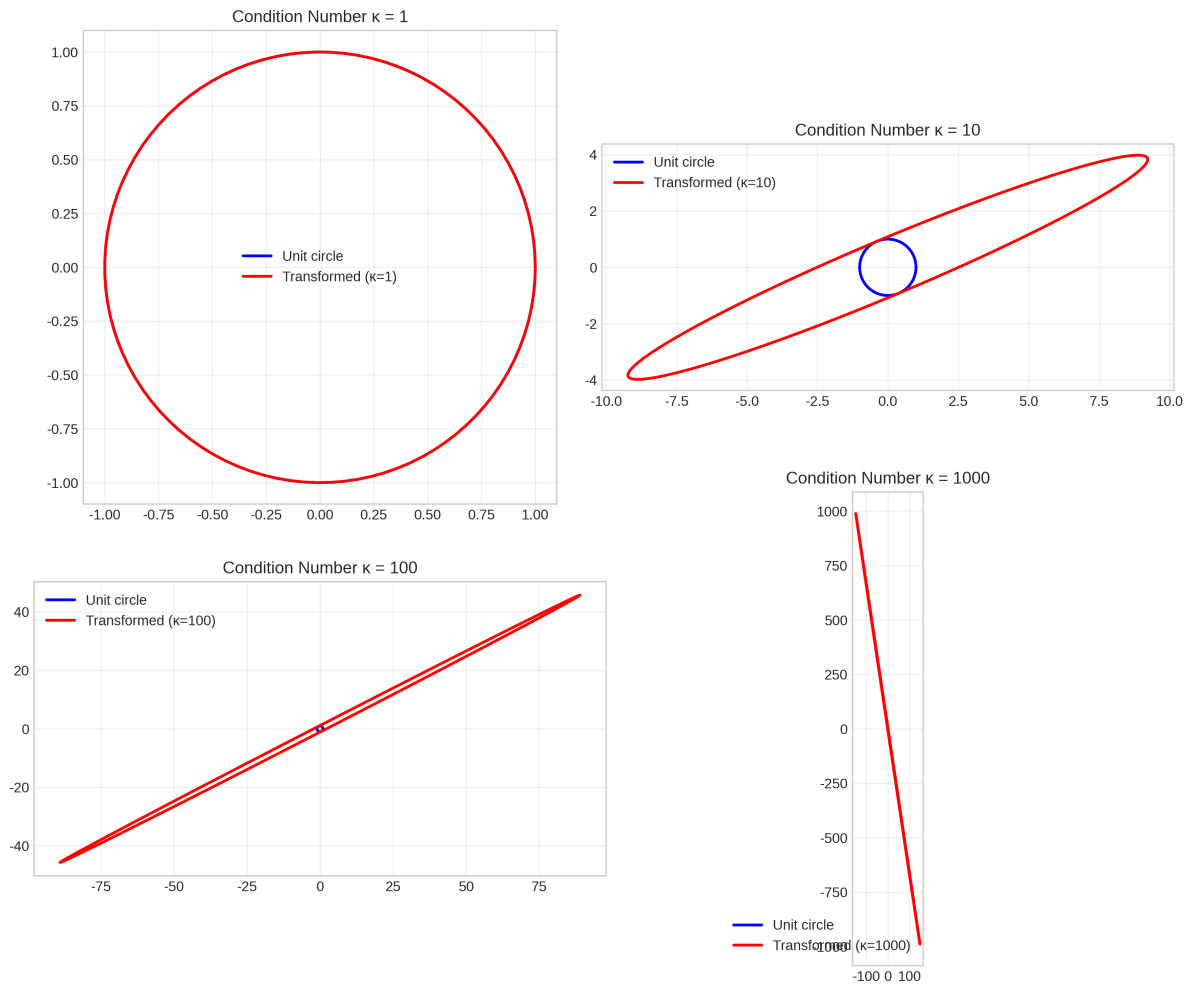

Figure 2: Matrix conditioning examples. Top: Well-conditioned vs ill-conditioned matrices and their effect on linear system solutions. Bottom: Condition number behavior for different matrix types including Hilbert matrices, random matrices, and structured matrices. The visualization shows how small perturbations in the right-hand side can lead to large changes in the solution for ill-conditioned systems.

2.2 Perturbation Analysis for Linear Systems

Theorem 2.2 (Perturbation Bound for Linear Systems)**.** *Consider the linear system $Ax = b$ where A is invertible. If we perturb the right-hand side to $b + \Delta b$, the solution changes from x to $x + \Delta x$ where $A(x + \Delta x) = b + \Delta b$.*

Then:

$$\frac{\|\Delta x\|}{\|x\|} \leq \kappa(A) \frac{\|\Delta b\|}{\|b\|}$$

This shows that the condition number bounds the relative error amplification.

Theorem 2.3 (Perturbation in the Matrix)**.** *If we perturb the matrix: $(A + \Delta A)(x + \Delta x) = b$, and $\|\Delta A\| < 1/\|A^{-1}\|$, then:*

$$\frac{\|\Delta x\|}{\|x + \Delta x\|} \leq \frac{\kappa(A)\|\Delta A\|/\|A\|}{1 - \kappa(A)\|\Delta A\|/\|A\|}$$

For small perturbations: $\frac{\|\Delta x\|}{\|x\|} \approx \kappa(A) \frac{\|\Delta A\|}{\|A\|}$

Example 2.1 (Hilbert Matrix Conditioning)**.** *The Hilbert matrix H_n with entries $h_{ij} = 1/(i+j-1)$ is notoriously ill-conditioned:*

For $n = 10$: $\kappa_2(H_{10}) \approx 1.6 \times 10^{13}$

This means that a relative perturbation of 10^{-16} in the data could result in a relative error of 10^{-3} in the solution, making the solution meaningless in double precision arithmetic.

Intuition: Geometric interpretation of conditioning

The condition number measures how "close" a matrix is to being singular:

- **Well-conditioned** ($\kappa \approx 1$): Matrix transformation preserves relative distances
- **Ill-conditioned** ($\kappa \gg 1$): Matrix transformation severely distorts the geometry
- **Singular** ($\kappa = \infty$): Matrix collapses the space to a lower dimension

The condition number quantifies the maximum distortion factor for any vector under the linear transformation.

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3 Special Classes of Matrices

3.1 Orthogonal Matrices

Definition. Orthogonal Matrix

A matrix $Q \in \mathbb{R}^{n\times n}$ is **orthogonal** if $Q^TQ = I$, equivalently $Q^T = Q^{-1}$.

For rectangular matrices $Q \in \mathbb{R}^{m\times n}$ with $m \geq n$, we say Q has **orthonormal columns** if $Q^TQ = I_n$.

Theorem 3.1 (Properties of Orthogonal Matrices)**.** *If Q is orthogonal, then:*

1. $\|Qx\|_2 = \|x\|_2$ *for all x (preserves Euclidean norm)*
2. $\det(Q) = \pm 1$
3. *All eigenvalues have absolute value 1*
4. $\kappa_2(Q) = 1$ *(perfectly conditioned)*
5. $(Qx)\cdot(Qy) = x\cdot y$ *for all x, y (preserves inner products)*

3.2 Symmetric Matrices

Definition. Symmetric Matrix

A matrix $A \in \mathbb{R}^{n\times n}$ is **symmetric** if $A = A^T$.

Theorem 3.2 (Spectral Theorem for Symmetric Matrices)**.** *If A is symmetric, then:*

1. *All eigenvalues are real*
2. *Eigenvectors corresponding to distinct eigenvalues are orthogonal*
3. *A can be diagonalized by an orthogonal matrix: $A = Q\Lambda Q^T$*
4. *A has an orthonormal basis of eigenvectors*

3.3 Positive Definite Matrices

Definition. Positive Definite Matrix

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is **positive definite** if $x^T A x > 0$ for all $x \neq 0$.

Equivalently:

- All eigenvalues are positive
- All leading principal minors are positive
- $A = B^T B$ for some invertible matrix B

Theorem 3.3 (Properties of Positive Definite Matrices)**.** *If A is positive definite, then:*

1. *A is invertible and A^{-1} is positive definite*
2. *$\kappa_2(A) = \lambda_{\max}/\lambda_{\min}$ where $\lambda_{\max}, \lambda_{\min}$ are the largest and smallest eigenvalues*
3. *The Cholesky decomposition $A = LL^T$ exists and is unique*
4. *$\|x\|_A = \sqrt{x^T A x}$ defines a norm on $\mathbb{R}^n$*

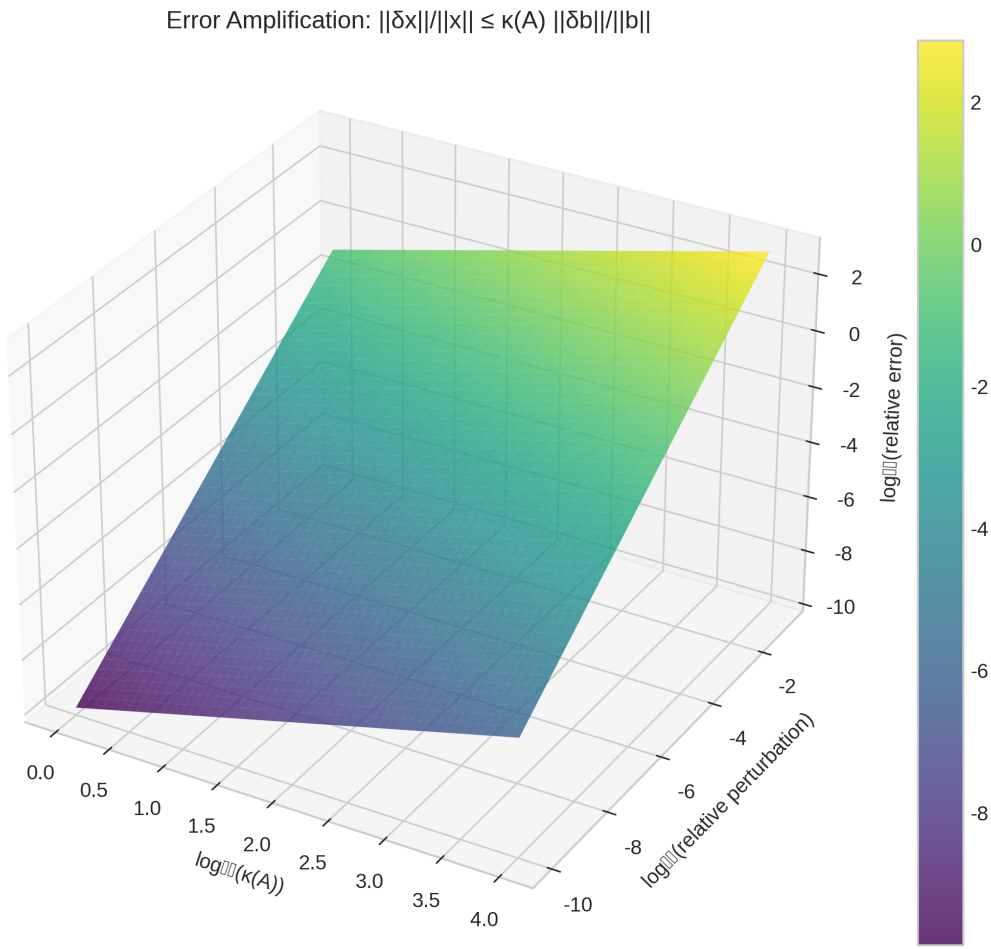

Figure 3: 3D visualization of matrix conditioning effects. The surface shows how the condition number varies as a function of matrix parameters, illustrating the relationship between matrix structure and numerical stability. Well-conditioned regions (blue) correspond to reliable numerical computation, while ill-conditioned regions (red) indicate potential numerical difficulties.

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4 Singular Value Decomposition

4.1 SVD Theorem and Properties

Theorem 4.1 (Singular Value Decomposition)**.** *For any matrix $A \in \mathbb{R}^{m\times n}$, there exist orthogonal matrices $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ such that:*

$$A = U\Sigma V^T$$

where $\Sigma \in \mathbb{R}^{m\times n}$ is diagonal with non-negative entries $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min(m,n)} \geq 0$.

*The σ_i are called the **singular values** of A.*

Definition. SVD Components

In the SVD $A = U\Sigma V^T$:

- **Left singular vectors:** columns of U, eigenvectors of AA^T
- **Right singular vectors:** columns of V, eigenvectors of A^TA

- **Singular values:** $\sigma_i = \sqrt{\lambda_i(A^T A)}$ where λ_i are eigenvalues

Theorem 4.2 (Fundamental Subspaces via SVD)**.** *Let $A = U\Sigma V^T$ with rank$(A) = r$. Then:*

- ***Range:*** $\mathcal{R}(A) = span\{u_1, u_2, \ldots, u_r\}$
- ***Null space:*** $\mathcal{N}(A) = span\{v_{r+1}, v_{r+2}, \ldots, v_n\}$
- ***Left null space:*** $\mathcal{N}(A^T) = span\{u_{r+1}, u_{r+2}, \ldots, u_m\}$
- ***Row space:*** $\mathcal{R}(A^T) = span\{v_1, v_2, \ldots, v_r\}$

4.2 Applications of SVD

Theorem 4.3 (Matrix Properties via SVD)**.** *For $A = U\Sigma V^T$:*

1. $\|A\|_2 = \sigma_1$ *(largest singular value)*
2. $\|A\|_F = \sqrt{\sum_{i=1}^{\min(m,n)} \sigma_i^2}$
3. *rank(A) = number of nonzero singular values*
4. *If A is square and invertible:* $\kappa_2(A) = \sigma_1/\sigma_n$

Definition. Moore-Penrose Pseudoinverse

For $A = U\Sigma V^T$ with rank r, the **pseudoinverse** is:

$$A^+ = V\Sigma^+ U^T$$

where Σ^+ is formed by transposing Σ and replacing each nonzero diagonal element σ_i with $1/\sigma_i$.

Properties:

- $AA^+A = A$ and $A^+AA^+ = A^+$
- $x^* = A^+b$ minimizes $\|Ax - b\|_2$ (least squares solution)
- Among all least squares solutions, x^* has minimum 2-norm

Example 4.1 (Low-Rank Approximation)**.** *The SVD provides the best low-rank approximation to a matrix. For $k < r =$ rank(A):*

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T$$

minimizes $\|A - B\|_2$ over all matrices B with rank$(B) \leq k$.

The approximation error is $\|A - A_k\|_2 = \sigma_{k+1}$.

Practice Problems

1. Prove that for any induced matrix norm, $\|A\| \geq \rho(A)$ where $\rho(A)$ is the spectral radius.
2. Show that the condition number satisfies $\kappa(A) \geq \|A\|\|A^{-1}\| \geq 1$.
3. For the 2×2 matrix $A = \begin{pmatrix} 1 & 1 \\ 0 & \epsilon \end{pmatrix}$, compute $\kappa_2(A)$ and analyze its behavior as $\epsilon \to 0$.

4. Prove that if A is symmetric positive definite, then $\kappa_2(A) = \lambda_{\max}/\lambda_{\min}$.

Solution:

Problem 1: For any eigenvalue λ of A with eigenvector v: $Av = \lambda v$. Taking norms: $|\lambda|\|v\| = \|\lambda v\| = \|Av\| \leq \|A\|\|v\|$. Thus $|\lambda| \leq \|A\|$ for all eigenvalues, so $\rho(A) = \max|\lambda| \leq \|A\|$.

Problem 2: Since $\|\cdot\|$ is an induced norm: $\|I\| = \|AA^{-1}\| \leq \|A\|\|A^{-1}\| = \kappa(A)$. Since $\|I\| = 1$, we have $\kappa(A) \geq 1$.

Problem 3: The singular values are $\sigma_1 = \sqrt{1 + 1/\epsilon^2 + 1}$ and $\sigma_2 = |\epsilon|$. Thus $\kappa_2(A) = \sigma_1/\sigma_2 \approx 1/|\epsilon|$ for small ϵ, showing the matrix becomes ill-conditioned as $\epsilon \to 0$.

Problem 4: For symmetric A, singular values equal absolute values of eigenvalues. Since A is positive definite, all eigenvalues are positive, so $\sigma_{\max} = \lambda_{\max}$ and $\sigma_{\min} = \lambda_{\min}$.

Summary and Looking Ahead

This week has established the mathematical foundations of numerical linear algebra, emphasizing the theoretical tools needed to analyze the behavior of computational algorithms. Key concepts include:

- Vector and matrix norms as measures of size and distance
- Condition numbers as indicators of problem sensitivity
- Special matrix classes and their computational advantages
- The SVD as a unifying tool for matrix analysis

These concepts provide the theoretical framework for understanding why certain linear algebra problems are computationally challenging while others are well-suited to numerical solution. The condition number, in particular, serves as a fundamental tool for predicting the reliability of numerical computations.

Next week, we will apply these theoretical insights to study direct methods for solving linear systems. We will see how matrix factorizations (LU, Cholesky, QR) exploit the structure of special matrix classes to provide efficient and stable algorithms. The conditioning concepts developed here will be essential for understanding when these methods succeed and when they may fail.